

UTTARADIT, Thailand

WMO: 483510

Lat: 17.6244N

Lon: 100.0969E

Elev: 64

StdP: 100.56

Time Zone: 7.00 (E07)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	15.8	17.2	10.4	7.9	22.5	12.0	8.8	22.7	3.7	27.3	3.2	26.9	0.1	10	0.310

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	11.4	38.7	26.1	37.5	26.0	36.4	25.9	27.8	34.1	27.5	33.6	27.2	33.1	1.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.4	22.1	30.3	26.1	21.6	30.1	25.8	21.2	29.8	89.4	34.4	88.0	33.8	86.7	33.2	31.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 3.4	2.7	2.1	13.5	40.2	1.9	1.3	12.2	41.1	11.1	41.9	10.0	42.6	8.7	43.6		
(5)			11.9	28.9	1.9	0.9	10.5	29.6	9.4	30.1	8.3	30.6	6.9	31.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	28.3	25.1	26.8	29.2	31.2	30.6	29.7	29.0	28.6	28.8	28.5	27.0	25.0	(6)
(7)		DBStd	2.54	1.94	1.88	1.86	1.80	2.08	1.40	1.39	1.22	1.08	1.28	1.77	2.21	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	1	1	0	0	0	0	0	0	0	0	0	0	1	(9)
(10)		CDD10.0	6684	468	472	596	637	639	592	589	577	564	573	509	466	(10)
(11)		CDD18.3	3643	211	238	338	387	381	342	331	319	314	314	259	208	(11)
(12)		CDH23.3	41568	1849	2479	4174	5370	4991	4182	3817	3494	3481	3422	2533	1776	(12)
(13)		CDH26.7	19653	768	1228	2293	3168	2706	1947	1589	1342	1386	1450	1082	695	(13)
(14)	Wind	WSAvg	0.3	0.2	0.3	0.3	0.4	0.3	0.3	0.2	0.2	0.2	0.2	0.3	(14)	
(15)	Precipitation	PrecAvg	1481	11	8	32	66	212	207	188	289	297	139	20	5	(15)
(16)		PrecMax	2100	125	39	130	191	399	418	332	481	554	252	72	24	(16)
(17)		PrecMin	1144	0	0	0	0	60	75	71	121	142	21	0	0	(17)
(18)		PrecStd	247	30	11	37	60	85	107	64	98	124	68	22	8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.1	36.3	38.5	40.5	40.4	37.6	36.1	35.0	34.9	35.0	34.9	34.0	(19)
(20)			MCWB	22.8	23.2	24.6	26.2	26.4	26.6	26.8	27.2	27.0	26.2	25.3	24.0	(20)
(21)		2%	DB	33.0	35.0	37.1	39.0	38.4	35.9	34.7	33.9	34.0	34.1	33.8	32.7	(21)
(22)			MCWB	22.6	23.3	24.7	26.2	26.6	26.8	26.8	26.9	26.9	26.0	24.8	23.3	(22)
(23)		5%	DB	32.0	34.1	36.1	37.8	37.0	34.6	33.7	32.9	33.2	33.4	32.9	31.7	(23)
(24)			MCWB	22.2	23.3	24.7	26.1	26.7	26.8	26.6	26.7	26.8	25.8	24.3	22.8	(24)
(25)		10%	DB	30.9	33.0	35.0	36.6	35.6	33.5	32.6	32.0	32.2	32.6	31.9	30.5	(25)
(26)			MCWB	21.9	23.1	24.7	26.1	26.6	26.7	26.4	26.6	26.6	25.6	24.0	22.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.1	26.1	27.0	28.1	28.3	28.1	27.7	27.8	27.9	27.7	26.7	25.7	(27)
(28)			MCD B	31.2	33.2	35.1	36.9	35.4	34.1	33.5	32.8	33.0	33.0	32.3	31.4	(28)
(29)		2%	WB	24.3	25.3	26.4	27.5	27.7	27.6	27.2	27.4	27.5	27.1	26.1	24.7	(29)
(30)			MCD B	30.3	32.4	34.4	36.0	34.8	33.4	32.6	32.2	32.3	32.1	31.4	30.1	(30)
(31)		5%	WB	23.6	24.7	25.9	27.0	27.3	27.2	26.9	27.1	27.1	26.7	25.6	23.9	(31)
(32)			MCD B	29.5	31.6	33.9	35.4	34.2	32.8	32.1	31.8	31.8	31.4	30.6	29.3	(32)
(33)		10%	WB	22.8	24.0	25.4	26.6	27.0	26.9	26.6	26.7	26.8	26.3	25.0	23.2	(33)
(34)			MCD B	28.5	30.9	33.2	34.7	33.7	32.2	31.4	31.0	31.2	30.8	29.7	28.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	12.5	13.3	12.5	11.4	9.3	7.7	7.0	6.8	7.3	8.4	9.9	11.1	(35)
(36)			MCDBR	13.7	14.3	13.7	12.7	11.4	9.4	8.7	8.2	8.5	9.5	10.9	11.8	(36)
(37)			MCWBR	4.8	5.1	4.5	3.7	3.0	2.7	2.6	2.8	2.8	2.9	3.4	3.9	(37)
(38)		5% WB	MCDBR	11.3	12.3	12.2	11.4	9.7	8.3	7.8	7.6	7.9	8.5	9.5	10.4	(38)
(39)			MCWBR	4.4	4.9	4.5	3.7	3.2	2.8	2.6	2.8	2.9	3.0	3.2	3.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.468	0.568	0.677	0.638	0.502	0.447	0.437	0.448	0.485	0.539	0.456	0.457	(40)	
(41)		taud	2.124	1.876	1.656	1.787	2.187	2.368	2.393	2.344	2.225	2.031	2.239	2.179	(41)	
(42)		Ebn at Noon	811	747	682	713	807	845	855	852	822	761	818	809	(42)	
(43)		Edh at Noon	149	200	257	226	150	124	121	128	143	169	132	137	(43)	
(44)	All-Sky Solar Radiation	RadAvg	4.64	5.19	5.41	5.97	5.49	4.93	4.59	4.49	4.62	4.71	4.81	4.47	(44)	
(45)		RadStd	0.25	0.22	0.31	0.31	0.46	0.38	0.33	0.27	0.25	0.27	0.24	0.20	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.19	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (22 neighbors)		+0.29	N/A	N/A	N/A	N/A	+0.12	N/A	N/A	+137	+139

Nomenclature: See separate page